



CO 3 - ASSESSMENT TOOL 2: Data Interpretation

COURSE NAME: FUNDAMENTALS OF DATA SCIENCE

COURSE CODE: DSA0404

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1. Data Summarization

A dataset of 11 employee salaries (in ₹'000) is: 32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120.

- (a) Compute the mean and median.
- (b) Explain why the mean is a poor measure of central tendency here.
- (c) Which measure of dispersion would you trust more — standard deviation or IQR — and why?

SOLUTION:

Dataset:

32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120

(a) Mean and Median

Mean:

$$\begin{aligned}\bar{x} &= \frac{32 + 35 + 36 + 38 + 40 + 42 + 45 + 48 + 50 + 95 + 120}{11} \\ &= \frac{581}{11} = \boxed{52.82}\end{aligned}$$

So, **Mean = ₹52.82 thousand**

Since there are 11 observations, the middle value is the 6th value:

So, **Median = ₹42 thousand**

(b) Why is mean a poor measure here?

The values **95 and 120** are unusually high compared with the other salaries. These extreme values pull the mean upward from the typical salary level.

- Mean = ₹52.82 thousand
- Median = ₹42 thousand

Therefore, the **median better represents the typical employee salary.**

(c) Standard deviation or IQR?

IQR is more trustworthy because the dataset contains extreme values/outliers.

- Standard deviation is strongly affected by extreme observations.
- IQR measures the spread of the middle 50% and is resistant to outliers.

Answer: IQR.

QUE:2.OutlierTreatment

Using the dataset above, apply the $1.5 \times \text{IQR}$ rule to identify outliers. Show Q1, Q3, IQR, and the lower/upper fences. Which values qualify as outliers? If you cap them at the fences (Winsorization) instead of removing them, how would the mean change qualitatively?

Sorted dataset:

32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120

Using the median-of-halves method:

Q1

Lower half:

32, 35, 36, 38, 40

Q1=36

Q3

Upper half:

45, 48, 50, 95, 120

Q3=50

IQR

$$IQR = Q3 - Q1$$

$$IQR = 50 - 36 = \boxed{14}$$

Lower Fence

$$\begin{aligned}
 &Q1 - 1.5(IQR) \\
 &= 36 - 1.5(14) \\
 &= 36 - 21 = \boxed{15}
 \end{aligned}$$

Upper Fence

$$\begin{aligned}
 &Q3 + 1.5(IQR) \\
 &= 50 + 1.5(14) \\
 &= 50 + 21 = \boxed{71}
 \end{aligned}$$

Outliers

Any value below 15 or above 71 is an outlier.

- $95 > 71 \rightarrow \text{Outlier}$
- $120 > 71 \rightarrow \text{Outlier}$

Therefore:

Outliers=95,120

Winsorization

If we cap the outliers at the upper fence of 71:

Original:

32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120

After capping:

32, 35, 36, 38, 40, 42, 45, 48, 50, 71, 71

New mean:

$$\frac{508}{11} = \boxed{46.18}$$

The mean changes from 52.82 to approximately 46.18.

Qualitatively: Winsorization reduces the influence of the extreme salaries, so the mean moves downward and closer to the median.

2. Measuring Asymmetry

Two datasets have the same mean (50) and standard deviation (10), but Dataset A has Pearson's skewness coefficient = +1.2 and Dataset B has -0.8. Sketch (describe) the shape of each distribution and state which is more likely to have salary-like data with a long right tail.

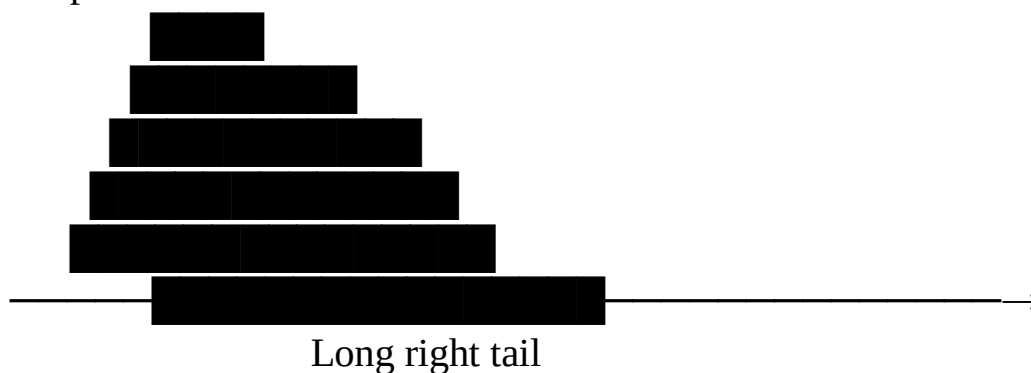
Both datasets have:

- Mean = 50
- Standard deviation = 10

Dataset A: Pearson's skewness = +1.2

A positive skewness means the distribution has a long tail toward the right.

Shape:



Most observations ← concentrated here long right tail

So Dataset A is right-skewed / positively skewed.

Dataset B: Pearson's skewness = -0.8

A negative skewness means the distribution has a long tail toward the left.

Shape:



Left tail

←—————| concentrated observations →

So Dataset B is left-skewed / negatively skewed.

Which is more like salary data?

Dataset A is more likely to represent salary data because salaries often have a long right tail: most employees earn moderate salaries while a smaller number of employees earn very high salaries.

3. Coefficient of Skewness Calculation

For a dataset: Mean = 62, Median = 58, Mode = 52, SD = 12.

Calculate skewness using

(a) Karl Pearson's coefficient (mode method),

(b) Karl Pearson's coefficient (median method, $Sk = 3(\text{Mean} - \text{Median})/\text{SD}$).

Compare the two results and comment on why they might differ slightly.

Given:

- Mean = 62
- Median = 58
- Mode = 52
- SD = 12

(a) Karl Pearson's coefficient — Mode method

Formula:

$$Sk = \frac{\text{Mean} - \text{Mode}}{SD}$$

Substitute:

$$Sk = \frac{62 - 52}{12}$$

$$= \frac{10}{12}$$

$$Sk = 0.833$$

Therefore, the distribution is **positively skewed**.

(b) Karl Pearson's coefficient — Median method

Formula:

$$Sk = \frac{3(\text{Mean} - \text{Median})}{SD}$$

Substitute:

$$Sk = \frac{3(62 - 58)}{12}$$

$$= \frac{3(4)}{12}$$

$$= \frac{12}{12}$$

$$Sk = 1.0$$

Comparison

Method	Skewness
Mode method	0.833
Median method	1.000

Both indicate **positive/right skewness**.

They differ slightly because the two formulas use **different measures of central tendency**—the mode in the first formula and the median in the second. For a moderately skewed distribution, these estimates need not be identical.

QUE: 5. Variance vs Standard Deviation Interpretation

Two production lines report these stats:

Line A: Mean = 200 units/day, Variance = 64

Line B: Mean = 150 units/day, Variance = 36

Which line has greater relative variability? Use the Coefficient of Variation to justify your answer, not just raw variance.

	Line A	Line B
Mean	200 units/day	150 units/day
Variance	64	36

First calculate standard deviation.

Line A

$$SD = \sqrt{64} = 8$$

Coefficient of Variation:

$$CV = \frac{SD}{Mean} \times 100$$

$$CV = \frac{8}{200} \times 100$$

$$CV = 4\%$$

Line B

$$SD = \sqrt{36} = 6$$

$$CV = \frac{6}{150} \times 100$$

$$CV = 4\%$$

Conclusion

Both production lines have the same relative variability:

$$CV_A = CV_B = 4\%$$

Although Line A has a higher raw variance (64 vs 36), it also has a higher mean (200 vs 150). Once variability is standardized relative to the mean using CV, both lines are equally variable.